Curriculum Vitae

Jason X. Wang Ph.D., MSc, BSc, FHEA

Lecturer in Sustainable Supply Chain Management

University of Sheffield

Staff profile: Jason X. Wang
Google Scholar - Jason X. Wang
ORCID (https://orcid.org/0000-0001-6807-5496)
Publon/Web of Science (Researcher ID: AAX-7454-2020)

Jason X. Wang

Lecturer in Sustainable Supply Chain Management
University of Sheffield
Sheffield Management School

Phone: +44 (0)114 222 3348
Email: jason.x.wang@sheffield.ac.uk

Education

Ph.D.	Operations & Supply Chain Management, University of Otago, New Zealand
MSc.	Operations & Supply Chain Management, University of Groningen, the Netherlands
BSc	International Economics & Business, University of Groningen, the Netherlands

Research Interests

Sustainable Supply Chain Management, Circular Economy, Blockchain Technology, Carbon Neutrality, Prefabricated Construction

Professional Appointments

2024–present	Lecturer in Sustainable Supply Chain Management, Sheffield Management School, University of Sheffield
2021–2024	MBA Program Leader (MBA/SLMDA), Huddersfield Business School, University of Huddersfield, United Kingdom
2021–2024	Senior Lecturer in Operations and Supply Chain Management, University of Huddersfield, United Kingdom
2019–2021	Lecturer in Operations and Supply Chain Management, University of Huddersfield, United Kingdom

Peer-reviewed Publications – Journal Publications

Dehe, B., **Wang, J. X.,** & Heath, S. (2026). The impact of blockchain technology on supply chain trust formation and development: The case of the pharmaceutical network. *Journal of Business Research*, *217*, 116473. https://doi.org/10.1016/j.jbusres.2026.116473	ABS: 3 ABDC: A JIF: 11.1
Wang, L., Xie, S., Na, A., **Wang, J. X.,** & Chien, F. (2026). The spillover effects of ESG performance on the trade credit: A supplier-customer dyad perspective. *Journal of Business Research*, *212*, 116218. https://doi.org/10.1016/j.jbusres.2026.116218	ABS: 3 ABDC: A JIF: 11.1
Wang, L., Meng, Y., Xiao, W., & **Wang, J. X.** (2026). Navigating the supply chain environmental sustainability: The moderating role of digitalisation in driving supply chain efficiency on circular economy performance. *Production Planning & Control*, *0*(0), 1–15. https://doi.org/10.1080/09537287.2026.2620478	ABS:3 ABDC: A JIF:5.4
Shi, Y., Venkatesh, V. G., Gao, Y., Gong, Y., **Wang, J. X**., Islam, S., & Luo, W. (2026). Exploring supply chain finance from the perspective of third-party logistics providers. *Production Planning & Control*, *37*(7), 710–726. https://doi.org/10.1080/09537287.2025.2579769	ABS:3 ABDC: A JIF:5.4
Xu, Y., Wang, L., **Wang, J. X.,** & Meng, X. (2025). Power dynamics and financial flexibility: Insights from supply chain dyads. *International Journal of Production Economics*, *289*, 109797. https://doi.org/10.1016/j.ijpe.2025.109797	ABS: 3 ABDC:A JIF:10.0

Wang, L., Xu, Y., **Wang, J. X**., & Shi, Y. (2025). Should Mining Enterprise Adopt Digital Transformation for Environmental Performance? A Supply Chain Power Structure Perspective. Business Strategy and the Environment, 34(2), 2469–2484. https://doi.org/10.1002/bse.4102 — ABS: 3 ABDC: A JIF: 12.5

Guan, P., Wood, L. C., **Wang, J. X.,** & Duong, L. N. K. (2024). Blockchain adoption in the port industry: A systematic literature review. *Cogent Business & Management*, *11*(1), 2431650. https://doi.org/10.1080/23311975.2024.2431650 — ABS=1 JIF: 3.0

Wang, J. X., Choi, T.-M., Wood, L. C., Olesen, K., & Reiners, T. (2023). When suppliers engage in sustainable supply chain management: How does the stock market react? *International Journal of Operations & Production Management. https://doi.org/10.1108/IJOPM-01-2023-0001* — ABS: 4 ABDC: A JIF: 7.4

Guan, P., Wood, L. C., **Wang, J. X**., & Duong, L. N. K. (2023). Barriers to blockchain adoption in the seaport industry: A fuzzy DEMATEL analysis. *Mathematical Biosciences and Engineering*, 20(12), 20995-21031. https://doi.org/10.3934/mbe.2023929 — JIF: 2.6

Zhang, A., Tay, H. L., Alvi, M. F., **Wang, J. X**., & Gong, Y. (2023). Carbon neutrality drivers and implications for firm performance and supply chain management. *Business Strategy and the Environment*, *32*(4), 1966–1980. https://doi.org/10.1002/bse.3230 — ABS:3 ABDC: A JIF:13.2

Zhang, A., Venkatesh, V. G., **Wang, J. X**., Mani, V., Wan, M., & Qu, T. (2023). Drivers of industry 4.0-enabled smart waste management in supply chain operations: A circular economy perspective in China. *Production Planning & Control*, *34*(10), 870–886.https://doi.org/10.1080/09537287.2021.1980909 — ABS:3 ABDC: A JIF:5.4

Burke, H., Zhang, A., & **Wang, J. X.** (2023). Integrating product design and supply chain management for a circular economy. *Production Planning & Control*, *34*(11), 1097–1113. https://doi.org/10.1080/09537287.2021.1983063 — ABS:3 ABDC: A JIF:5.4

Zhang, A., Alvi, M. F., Gong, Y., & **Wang, J. X.** (2022). Overcoming barriers to supply chain decarbonization: Case studies of first movers. *Resources, Conservation and Recycling*, *186*, 106536. https://doi.org/10.1016/j.resconrec.2022.106536 — JIF:13.2

Wood, L. C., Duong, L. N. K., & **Wang, J. X**. (2022). Business process improvement for sustainable technologies investments in construction: A configurational approach. *Sustainability*, *14*(9), 5697. https://doi.org/10.3390/su14095697 — JIF:3.9

Wang, J. X., Burke, H., & Zhang, A. (2022). Overcoming barriers to circular product design. *International Journal of Production Economics*, 243, 108346. https://doi.org/10.1016/j.ijpe.2021.108346 — ABS: 3 ABDC: A JIF: 10.0

Zhang, A., **Wang, J. X.,** Farooque, M., Wang, Y., & Choi, T.-M. (2021). Multi-dimensional circular supply chain management: A comparative review of the state-of-the-art practices and research. *Transportation Research Part E: Logistics and Transportation Review*, *155*, 102509. https://doi.org/10.1016/j.tre.2021.102509 — ABS: 3 ABDC: A* JIF: 9.3

Duong, L. N. K., **Wang, J. X.,** Wood, L. C. Torsten R., & Kourshan M. (2021). The value of incremental environmental sustainability innovations in the construction industry: An event study. *Construction Management and Economics,* 39(5), 398–418. https://doi.org/10.1080/01446193.2021.1901950 — ABS: 2 ABDC: A JIF:3.4

Wood, L. C., **Wang, J. X.**, Duong, L. N. K., Reiners, T., & Smith, R. (2018). Stock market reactions to auto manufacturers' environmental failures. *Journal of Macromarketing*,38(4), 364–382. https://doi.org/10.1177/0276146718781915 — ABS: 2 ABDC: A JIF: 3.1

Wood, L. C., & **Wang, J. X.** (2018). The event study method in logistics research: Overview and a critical analysis. *International Journal of Applied Logistics*, *8*(1), 57–79. https://doi.org/10.4018/IJAL.2018010104

Wood, L. C., **Wang, J. X.**, Olesen, K., & Reiners, T. (2017). The effect of slack, diversification, and time to recall on stock market reaction to toy recalls. *International Journal of Production Economics*, 193, 244–258, https://doi.org/10.1016/j.ijpe.2017.07.021 — ABS: 3 ABDC: A JIF: 10.0

Duong, L. N. K., Wood, L. C., **Wang, J. X.,** & Wang, W. Y. C. (2017). Transport on-demand in a service supply chain experiencing seasonal demand: Managing persistent backlogs. *International Journal of Operations Research*, *14*(3), 121–138,http://www.orstw.org.tw/ijor/vol14no3/IJOR2017_vol14_no3_p121_p138.pdf

Peer-reviewed Publications – Editorials

Wang, L., Xiong, Y., Wu, J., **Wang, J. X**, & Lam, H. K. S. (2026). Editorial: Web 3.0 and Metaverse technologies in logistics and supply chain management. *Transportation Research Part E: Logistics and Transportation Review*, 214, 105037. https://doi.org/10.1016/j.tre.2026.105037 — ABS: 3 ABDC: A* JIF: 9.3

Peer-reviewed Publications – Working Papers

Wang, J.X. (Forthcoming). 'Blockchain for Circular Economy'. In A. Genovese, A. Marotti de Mello, & B. Lowe (Eds.), *Dictionary of Circular Economy*. Cheltenham: Edward Elgar Publishing.

Wang, J. X., Zhang, W., Zhang, A., Shi, J., & Gong, Y. (Under Revision). Investor Optimism, Public Skepticism: Divergent Stakeholder Responses to Carbon Neutrality Achievement and Emissions Trading. *Production and Operations Management*. ABS: 4 ABDC: A* JIF: 5.1

Wang, J. X., Ma, J., Zhang, A., Wood, L. C., & Abualqumboz, M. (Under revision). Untangling the Productivity Puzzle in Offsite Construction Supply Chains: A Theory of Swift, Even Flow Perspective. *International Journal of Operations & Production Management*. ABS: 4 ABDC: A JIF: 7.4

Peer-reviewed Publications - Book Chapters

Dehe, B., Mirzaei, M., **Wang, J. X.,** & Wood, L. C. (2026). An ecosystem of innovation: The transforming role of digitalisation and sustainability in buyer–supplier relationships. In Research Handbook on Buyer–Supplier Relationships (pp. 168–183). Edward Elgar Publishing. https://www.elgaronline.com/edcollchap/book/9781035321025/chapter11.xml

Wang, J. X., Zhang, W., & Ma, J. (2024). Unlocking doing well by doing good: A sustainable and ethical operations perspective. In T.-M. Choi (Ed.), Encyclopedia of Operations Management. Elsevier.https://doi.org/10.1016/B978-0-443-28993-4.00014-7

Wang, J. X. (2018). The effect of supply chain sustainability management in the perspective of suppliers. In L. C. Wood (Ed.), *Contemporary Approaches and Strategies for Applied Logistics* (pp. 77–100). Hershey, PA: IGI Global. https://doi.org/10.4018/978-1-5225-5273-4.ch004

Wang, X., & Wood, L. C. (2016). The influence of supply chain sustainability practices on suppliers. In B. Christiansen (Ed.), *Handbook of Research on Global Supply Chain Management* (pp. 531-544). Hershey, PA: IGI Global. https://doi.org/10.4018/978-1-7998-0945-6.ch083

Duong, N. K. L., Wood, L. C., & **Wang, X.** (2016). Review of RFID applications in perishable inventory management. In B. Christiansen (Ed.), *Handbook of Research on Global Supply Chain Management* (pp. 139-146). Hershey, PA: IGI Global. https://doi.org/10.4018/978-1-4666-9639-6.ch008

Peer-reviewed Publications - Conference Proceedings

Wang, J. X., Ma, J., (2026). The Supply Chain Perspective of manufacturing innovation: An offsite construction context. *24th International Working Seminar on Production Economics*, 23-27 February, 2026, Innsbruck, Austria.

Wang, J. X., Ma, J., (2025). When does it pay to do off-site construction? A supply chain management perspective. *2025 European Operations Management Association (EurOMA) Annual Conference*, 13-18 June, 2025, Milan, Italy.

Heath, S., Dehe, B., & **Wang, J. X.** (2023). The impact of blockchain technology on the nature of trust: A pharmaceutical supply chain perspective. *Operations and Supply Chain Management Conference and 1st Symposium on Blockchain Research*, 28-30 June, 2023, Liverpool, United Kingdom.

Zhang, A., **Wang, J. X.,** Farooque, M., Wang, Y., & Choi, T.-M. (2021). Multi-dimensional circular supply chain management: A comparative review of the state-of-the-art practices and research. *2021 International Conference on Resource Sustainability,* 19-23 July, Dublin, Ireland.

Wang, J. X., Wood, C. L., Olesen, K., & Reiners, T. (2019). The effect of the consolidated supply chain sustainability management of buyers on the market value of suppliers. *Production and Operations Management Society (POMS) 2019 International Conference*, 2-4 September, 2019, Brighton, United Kingdom

Smith, R., Wood, L. C., & **Wang, J. X.** (2017). Caught when failing to meet environmental standards: The case of the automotive industry. *The 15th ANZAM Operations, Supply Chain and Services Management Symposium*, 13-14 June, 2017, Queenstown, New Zealand

Wood, L. C., Smith, R., & **Wang, J. X.** (2017). Failure to meet marketed environmental standards: The case of automobiles. *The 42nd Annual Conference of the Macromarketing Society*, 19-23 June, 2017, Queenstown, New Zealand

Wang, X., Wood, C. L., & Olesen, K. (2016). The impact of customer firms' supply chain sustainability management practices on suppliers' financial performance. *The 25th Annual Conference of the International Purchasing and Supply Education and Research Association (IPSERA)*, 20-23 March, 2016, Dortmund, Germany

Duong, L. N. K., Wood, L. C., **Wang, X.**, & Wang, W. Y. C. (2016). Information accuracy and KPIs for service subcontractors. *The 2016 Manufacturing and Service Operations Management Conference (MSOM)*, 30th June - 1st July, Auckland, New Zealand

Duong, N. K. L., Wood, L. C., **Wang, J.X.,** Wang, W. Y. C. (2016). Persistent backlogs: Improving service performance at a logistics service company. *The 6th International Conference on Transportation and Logistics,* 7–9 September, 2016, Taiwan

Chong, J. L. L., Nagar, S., & **Wang, X.** (2015). Market reactions to enterprise systems investments in European economy: An event study. *The 12th European, Mediterranean & Middle Eastern Conference on Information Systems*, 1 – 2 June, 2015, Athens, Greece

Wood, C. L., Wood, A., Reiners, T., Duong, L. N. K., & **Wang, X.** (2014). An exploration on the New Zealand use of technology to facilitate logistics. *The 25th Australasian Conference on Information Systems*, 8-10 December, 2014, Auckland, New Zealand

Wang, X., Petkova, B. N., & Wood, C. L. (2014). Do suppliers benefit from supply chain sustainability programs? --- The case of Wal-Mart. *The 12th ANZAM Operations, Supply Chain and Services Management Symposium*, 3-4 July, 2014, Auckland, New Zealand

Industry Magazine (non-refereed)

Wang, X., Petkova, B. N., & Wood, C. L. (2015). Do suppliers benefit from supply chain sustainability programs? The case of Wal-Mart. *LOGI-BIZ,* 15(8), 66-69. http://www.logi-biz.com/backnumber/2015_11.html

Professional Services

Editorial Board

Department Editor (Operations Research, Information and Technology) *Cogent Business & Management* (Impact Factor:3.0, ABS:1, Taylor & Francis), 2022 - present

Editorial Review Board Member, *Journal of Supply Chain Management* (Impact Factor:10.6, ABS:4, ABDC: A, Wiley), 2021 - present

Editorial Review Board Member, *IEEE Transactions on Engineering Management* (Impact Factor: 4.6, ABS:3, ABDC: A, IEEE Xplore), 2025 - present

Editorial Review Board Member, *International Journal of Operations & Production Management* (JIF: 7.4, ABS: 4, ABDC: A, Emerald), 2026 - present

Editorial Board Member, Operations Management, (MDPI), 2026 – present

Special Issue Guest Editor

"Editorial: Web 3.0 and Metaverse technologies in logistics and supply chain management," *Transportation Research Part E: Logistics and Transportation Review (Impact factor* = 9.3, ABDC: A*, ABS: 3)

Conference Contributions

2027 International Journal of Operations & Production Management (IJOPM) Workshop: Developing the Next Generation of OSCM Scholarship, 4th -5th January 2027, Hangzhou, China (Hosted by School of Management, Zhejiang University)	Organizing Committee Member, Mentoring Session Mentor & Parallel Session Chair
4th Operations and Supply Chain Management Symposium: Emerging Technologies and the Industry 5.0, 1st – 2nd September 2026, Sheffield, UK (Hosted by Sheffield University Management School)	Organizing Committee Chair
IEEE TEMSCON Global 2025 Conference, 4th – 7th August 2025, San Diego, CA, USA (IEEE Technology and Engineering Management Society)	Program Committee Member

Grant/Funding Review

Invited expert assessor (November, 2026), the Dutch Research Council (NWO) – Invited to assess grant applications for the 'KIC Demand-driven Research By Consortia S – 2026' call.

Journal Reviewers (Publon/Web of Science)

Production and Operations Management
Journal of Operations Management
Journal of Supply Chain Management
International Journal of Operations and Production Management
Transportation Research Part E: Logistics and Transportation Review
IEEE: Transactions on Engineering Management
International Journal of Production Research

Teaching Experience

Ph.D. Supervision

Retno Setiyawati, 2027 Feb – present, Primary Supervisor, University of Sheffield	A Dynamic Capability-Based Model for Sustainable Supply Chain Transformation
Peng Guan, 2022-2025, Co-supervisor, University of Otago	The application of blockchain technology in port management
Dina Tamazin, 2020–2024, Co-supervisor, University of Huddersfield	Supply chain performance measurement model for oil refining companies: A sustainability perspective

Ph.D./DBA Examination

2020 (Ph.D.): Emanuel Vincent Farrugia	*An investigation of Malta's evolution in global value chains and the implications for sustainable development*, Manchester Metropolitan University
2021(Ph.D.): Walid Mourhrib	*Adoption of the Physical Internet (PI): Key Implementation Challenges*, University of Huddersfield
2023 (DBA): Svenja Köster	*Open innovation through supplier management - An approach to leverage the innovation potential for financial service companies*, Northumbria University
2023 (Ph.D.): Philip Chilmon	*Sustainable Public Procurement - Conceptualization, Integration and Application*, University of Huddersfield

Taught modules

University of Sheffield

MGT6150 Operations Management (MBA), 24-25, 25-26, Module leader

MGT6085 Global Supply Chain Leadership (MSc Logistics and Supply Chain Management), 24-25, 25-26 Lecturer

MGT6155 Supply Networks Management/Logistics System-Advanced Optimization (MSc Logistics and Supply Chain Management), 25-26, Lecturer

New module development/Module Leader – MGT422 Advanced Analytics: Econometrics and Forecasting (MSc Business Analytics), 26-27 (responsible for the new module development on learning outcomes, teaching & learning topics, teaching team management, delivery methods/materials, assessment strategy)

New module development/Module Leader - Supply Chain Networks - Strategy and Operations (MBA), 27-28 (responsible for the new module development on learning outcomes, teaching & learning topics, teaching team management, delivery methods/materials, assessment strategy, and response/revision to the review by Faculty Academic Programmes Office)

University of Huddersfield

BMR1012 Operations Management (MSc Project Management and Operations Management), 19-20, 20-21, 21-22, 22-23, 23-24, Module leader, University of Huddersfield

BMT1070 Purchasing and Supply Chain Management (MSc Logistics and Supply Chain Management; MSc Sustainable Supply Chain Management, MSc Supply Chain Management with Humanitarian Challenges), 20-21, 21-22, 22-23, 23-24, Module Leader, University of Huddersfield

BMS0069 Business Research Project (MBA), 21-22, 22-23, 23-24, Module Leader, University of Huddersfield

BMS0071 Work-based Project (Senior Leader Master's Degree Apprenticeship/MBA), 21-22, Module Leader, University of Huddersfield

BMT 1018 Dissertation (MSc Logistics and Supply Chain Management & MSc Operations and Supply Chain Management), 2018-19, Module Leader, University of Huddersfield

BMR1003 Essential of Operations Management (MSc Operations and Supply Chain Management), 2018-19, Module Leader, University of Huddersfield

Logistics and Supply Chain Management (MBA Aviation Management), 2019, 2020, 2021, 2022, 2023 Lecturer, University of Huddersfield

BHT4024 Supply Chain Management – Work-based Development 3 (BA [Hons] Business with Supply Chain Management Professional [Apprenticeship Program]), 2020-21, Module Leader, University of Huddersfield

BIT2018 Purchasing and Supply (BSc [Hons] Logistics & Supply Chain Management), 2018-19, Lecturer; 2019-20, 2020-21, Module Leader, University of Huddersfield

BIR2001 Operations Management (BA [Hons] Business Studies), 2018-19, Lecturer, 2019-20, Module Leader, University of Huddersfield

University Services

University of Sheffield

Deputy Chair (University of Sheffield) in *N8 Operations and Supply Chain Management Research Seminar Series (*Research partnership of the eight most research intensive Universities in the North of England-*https://www.n8research.org.uk/)*

MBA selection panelist, 2025-present, responsible for MBA recruitment activities, MBA candidates' interviews, and offers

Academic representative in MBA overseas recruitment, March-April 2025, April 2026, representing Sheffield University Management School in the 'QS Discover MBA' in Tokyo, Japan, and Taipei, Taiwan.

University of Huddersfield

MBA program leader (Master of Business Administration/Senior Leader Master's Degree Apprenticeship[SLMDA]), 2021-2024

Committee member in Huddersfield Business School Accreditation and Validation Panel (SAVP), 2020-2024

PhD coordinator in the Department of Accounting, Finance, Logistics and Economics, 2020-2024

Coordinator in BA (Hons) Business with Supply Chain Management Professional, Chartered Manager Degree Apprenticeship (CMDA), 2021–2021

Project member (Mathematics route) in Partnership for Innovation of Exact Supporting Courses in Economics and Management Study Programmes (Funded by European Commission with Erasmus+ KA2 – Cooperation for Innovation and the Exchange of Good Practices, KA203 – Strategic Partnerships for Higher Education), 2021-2022

Organizing committee member of Huddersfield Business School Research Conference 2020 and track chair in Emerging Economies and Globalization

University academic integrity officer, 2019-2024

University of Reading, Henley Business School

External Examiner for Executive MBA Global/Flexible/Levy Executive MBA (Overseeing modules of Managing Processes and Systems and Managing the Project-based Environment across multiple international teaching locations, including the UK, Nordics, South Africa, and Malta), 2026-present

Northumbria University of Newcastle, Newcastle Business School

External Examiner for Operations, Supply Chain and Logistics Management Subject Group, Department of Marketing, Operations, and Systems (MOS), 2022–2026

Grants and Funded Projects (approximately GBP 280,000 in total)

2026 Co-invetigator, "Research on Supply Chain Risk Propagation and Response Strategies for Critical Components in High-End Equipment Manufacturing" (面向高端装备制造业关键零部件的供应链风险传染与响应策略研究), CNY 400,000 / ~£43,000, National Natural Science Foundation of China, 1st January, 2026 – 31st December, 2029, Grant number: 72571027

2023 Co-investigator, "Investigating management decision making and readiness in SMEs – how technology-driven development affects long term productivity growth of logistics operations," The Productivity Institute, GBP 16,000, Strategic productivity for business functions and leadership teams, Economic and Social Research Council (ESRC)/The Productivity Institute https://www.ukri.org/opportunity/strategic-productivity-for-business-functions-and-leadership-teams/

2021 Co-investigator, "Cloud-based dynamic end-to-end control tower for supply chain, a decision support system," the University of Huddersfield, Area of Strategic Research Importance (ASRI) Project, GBP 250,321.

2020 Principal Investigator, "Sustainable innovation in manufacturing: An offsite construction perspective," Huddersfield Business School Research Fund, GBP 3,000.

2022 Principal Investigator, "Sustainable Operations and Supply Chain Management," University of Huddersfield, International Collaboration Research Projects with University of Otago, New Zealand, GBP 3,700.

2015 Co-investigator, "Managing persistent backlogs: Mitigating service failure," NZD 15,000 (approximately GBP 7200), A1 Onehunga Towing Services Ltd and Auckland University of Technology, New Zealand.

Industrial Engagement

Industrial Advisory and Knowledge Transfer - Sustainable Procurement for Event: The Circular Economy and Responsible Resources Management, 2021, *providing training for the procurement managers in a world-leading electric motorsport company on the circular economy concept and its practices in supply chains*

Industrial Consulting - Ad: Venture - Elevate Programme, 2021 Consulting program for local start-up businesses funded by the European Union and Kirklees Council, The United Kingdom, *providing mentoring and access to specialist advice and support in the areas of supply chain data analytics*

Industrial Consulting - Managing Persistent Backlogs: Mitigating Service Failure, 2015–2016, funded applied research project, *providing a consulting service concerning operational failure in a vehicle recovery service supply chain.*

Membership

IEEE Technology and Engineering Management Society (TEMS)

Fellowship of The Higher Education Academy (HEA)